

6.5

- (a) Let $X_1, X_2, \dots, X_9$ be a random sample of the number of hours 10-year-old boys play video games in a week. Then

$$\begin{aligned} P(8 < \bar{X} < 12) &= P\left(\frac{8 - 10}{\frac{4}{\sqrt{9}}} < Z < \frac{12 - 10}{\frac{4}{\sqrt{9}}}\right) = P(-1.5 < Z < 1.5) \\ &= \Phi(1.5) - \Phi(-1.5) = 0.9332 - 0.0668 = 0.8664 \end{aligned}$$

- (b) Let X be the number of hours a 10-year-old boy plays video games in a week. Then

$$\begin{aligned} P(X > 16) &= P\left(Z < \frac{16 - 10}{4}\right) = P(Z > 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 \\ &= 0.0668 \end{aligned}$$

Let Y be the number of 10-year-old boys who are addicted to video games. Then $Y \sim \text{Bin}(12, 0.0668)$.

$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - 0.9332^{12} = 1 - 0.4362 = 0.5638$$

6.8

Let $X_1, X_2, \dots, X_{60}$ be a random sample from $N(5, 25)$. Let $Y = \sum_{i=1}^{60} X_i$. Then $Y \sim N(300, 1500)$.

- (a)

$$P(Y < 300) = P\left(Z < \frac{300 - 300}{\sqrt{1500}}\right) = \Phi(0) = 0.5$$

- (b)

$$P(Y > 390) = P\left(Z > \frac{390 - 300}{\sqrt{1500}}\right) = 1 - \Phi(2.324) = 1 - 0.9899 = 0.0101$$

6.14

- (a) Let X be the face value in a toss. Then $f(x) = \frac{1}{4}, x = 1, 2, 3, 4$.

$$E(X) = \sum_{x=1}^4 xf(x) = \frac{1+2+3+4}{4} = 2.5$$

$$E(\bar{X}) = E(X) = 2.5$$

- (b)

$$E(X^2) = \sum_{x=1}^4 x^2 f(x) = \frac{1+4+9+16}{4} = 7.5$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 7.5 - 2.5^2 = 1.25$$

$$\sigma = \sqrt{1.25} = 1.118$$

- (c)

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{1.118}{\sqrt{200}} = 0.079$$

- (d) By CLT, $\bar{X}$ is approximately $N(2.5, 0.079^2)$.

$$P(\bar{X} < 2.7) \approx P\left(Z < \frac{2.7 - 2.5}{0.079}\right) = \Phi(2.53) = 0.9943$$

6.16

- (a) $\bar{X} \sim N(\mu, \sigma^2/n) = N(69.5, 9/20) = N(69.5, 0.45)$

$$\begin{aligned} P(68 < \bar{X} < 70) &= P\left(\frac{68 - 69.5}{\sqrt{0.45}} < Z < \frac{70 - 69.5}{\sqrt{0.45}}\right) \\ &= P(-2.236 < Z < 0.745) = \Phi(0.745) - \Phi(-2.236) \\ &= 0.7719 - 0.0127 = 0.7592 \end{aligned}$$

- (b) $\bar{X} \sim N(\mu, \sigma^2/n) = N(69.5, 9/n)$ for a large n .

$$\begin{aligned} P(68.52 < \bar{X} < 70.48) &= P\left(\frac{68.52 - 69.5}{\frac{3}{\sqrt{n}}} < Z < \frac{70.48 - 69.5}{\frac{3}{\sqrt{n}}}\right) \\ &= P\left(-\frac{0.98\sqrt{n}}{3} < Z < \frac{0.98\sqrt{n}}{3}\right) = 1 - 2P\left(Z > \frac{0.98\sqrt{n}}{3}\right) \\ &= 0.95 \Rightarrow P\left(Z > \frac{0.98\sqrt{n}}{3}\right) = 0.025 \Rightarrow \Phi\left(\frac{0.98\sqrt{n}}{3}\right) = 0.975 \\ &\Rightarrow \frac{0.98\sqrt{n}}{3} = 1.96 \Rightarrow \sqrt{n} = \frac{3(1.96)}{0.98} = 6 \Rightarrow n = 36 \end{aligned}$$

(c) Let $X \sim N(69.5, 9)$. Then

$$P(X < 68) = P\left(Z < \frac{68 - 69.5}{3}\right) = \Phi(-0.5) = 0.3085.$$

Let Y be the number of American men who are shorter than 68 inches. Then $Y \sim \text{Bin}(100, 0.3085)$. $\mu = np = 100(0.3085) = 30.85$, $n(1 - p) = 100(0.6915) = 69.15$ are large and $\sigma = \sqrt{np(1 - p)} = \sqrt{100(0.3085)(0.6915)} = 4.6187$.
 $P(Y \geq 25) \approx P\left(Z \geq \frac{24.5 - 30.85}{4.6187}\right) = P(Z \geq -1.375) = \Phi(1.375) = 0.9154$

6.18

(a)

$$E(X_1) = \int_{-\infty}^{\infty} xf(x)dx = \int_0^1 \left(\frac{3x^3}{2} + x^2\right)dx = \left[\frac{3x^4}{8} + \frac{x^3}{3}\right]_0^1 = \frac{3}{8} + \frac{1}{3} = \frac{17}{24}$$

(b)

$$E(X_1^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^1 \left(\frac{3x^4}{2} + x^3\right)dx = \left[\frac{3x^5}{10} + \frac{x^4}{4}\right]_0^1 = \frac{3}{10} + \frac{1}{4} = \frac{11}{20}$$

$$\text{Var}(X_1) = E(X_1^2) - [E(X_1)]^2 = \frac{11}{20} - \left(\frac{17}{24}\right)^2 = \frac{139}{2880} = 0.0483$$

(c) By CLT, $\bar{X}$ is approximately

$$N(\mu, \sigma^2/n) = N\left(\frac{17}{24}, (0.0483)/_{100}\right) = \left(\frac{17}{24}, 0.000483\right).$$

$$\begin{aligned} P(0.7 < \bar{X} < 0.75) &\approx P\left(\frac{0.7 - 0.708}{\sqrt{0.000483}} < Z < \frac{0.75 - 0.708}{\sqrt{0.000483}}\right) \\ &= P(-0.36 < Z < 1.91) = \Phi(1.91) - \Phi(-0.36) \\ &= 0.9719 - 0.3594 = 0.6125 \end{aligned}$$

6.25

$$\chi_{0.05,14}^2 = 23.6848$$

$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{14S^2}{3}$ is distributed as χ^2 with 14 degrees of freedom.

(a)

$$0.05 = P\left(\frac{14S^2}{3} > 23.6848\right) = P\left(S^2 > \frac{3(23.6848)}{14}\right) = P(S^2 > 5.0753)$$

$$P(S^2 < 5.0753) = 0.95$$

The 95th percentile is 5.0753.

(b)

$$>3*\text{qchisq}(0.95, 14)/14$$

$$5.075312$$

7.1

(a)

$$\bar{x} = \frac{10 + 8 + 9 + 7 + 11 + 13}{6} = \frac{58}{6} = 9.667$$

(b)

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2/n}{n-1} = \frac{584 - 58^2/6}{5} = 4.667$$

$$s = \sqrt{4.667} = 2.16$$

(c) $\text{se}(\bar{X}) = \sigma_{\bar{X}} = \sigma/\sqrt{n}$

An estimate of the standard error is

$$\frac{s}{\sqrt{n}} = \frac{2.16}{\sqrt{6}} = 0.882$$

7.2

$$x = 55, n = 100$$

(a)

$$\hat{p} = \frac{x}{n} = \frac{55}{100} = 0.55$$

(b)

$$\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$$

Standard error estimate is

$$\sqrt{\hat{p}(1-\hat{p})/n} = \sqrt{(0.55)(0.45)/100} = 0.05$$